

Reading a district from open map data

Eixample, Barcelona · 7.51 km² · Overture Maps 2026-08-19.0

What can a planner honestly read from Overture alone — and what can they not?

Overture is the **only** source: no OSM top-up, no municipal open data, no GTFS. Where Overture is thin, that is a **finding**, not a hole to patch.

No socioeconomic data anywhere — not as a KPI, not as a denominator. Denominators are street length, segment count or building count.

Five KPIs: a **share**, two **densities**, a **distance**, and a second share. Draw any polygon; every number recomputes for that polygon alone.

Why Eixample

7.51 km², a grid dense enough that a drawn block still has real sample sizes, and a place with a **known intervention** to test against: the Sant Antoni superblock.

The district read

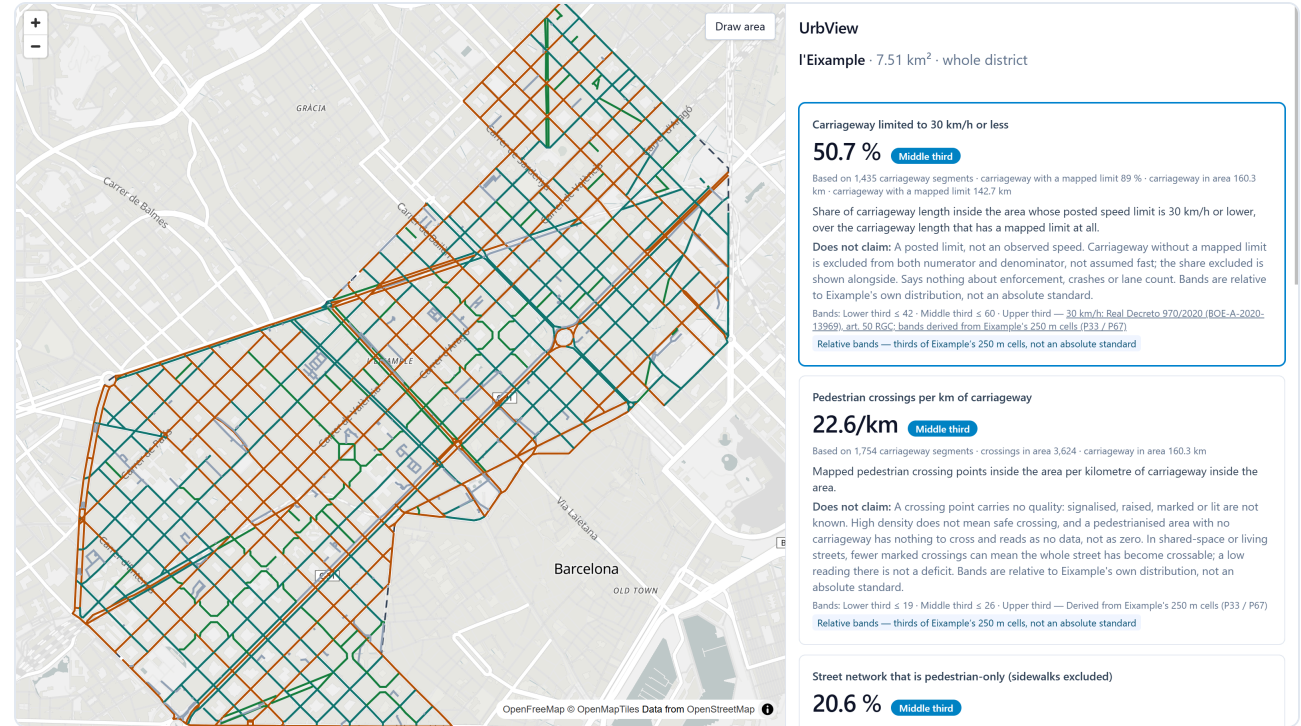
50.7 % carriageway limited to 30 km/h
(89 % has a mapped limit)

22.6 crossings per km of carriageway

20.6 % street network that is pedestrian-only

362 m median distance to a public green ≥ 0.5 ha

55.4 street trees per km of
carriageway



Only 41 % of Eixample's 8,397 buildings are within 300 m of a public green space of at least 0.5 ha. The WHO rule of thumb says they should be.

The district reads "Middle third" on the four relative bands by construction — it *is* the distribution they were cut from.

Two areas, same size

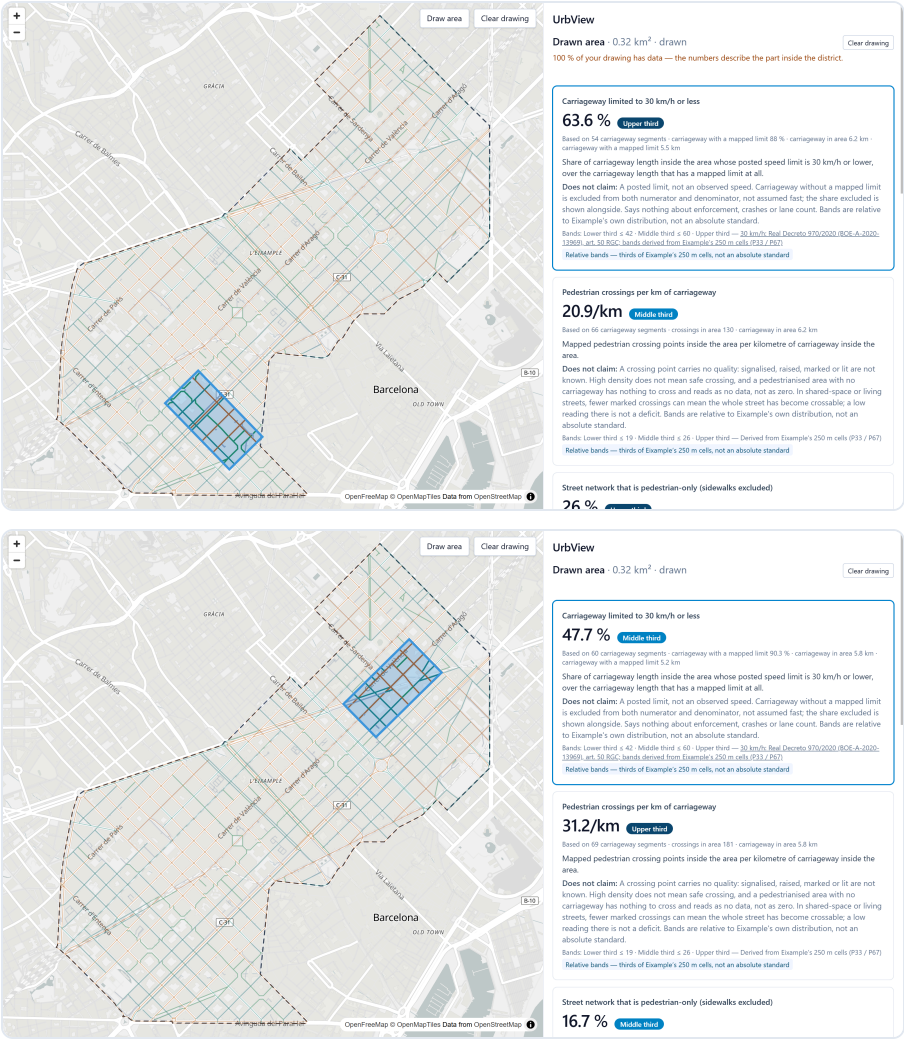
A — Sant Antoni superblock

Comte Borrell, Tamarit → Consell de Cent
0.318 km²

B — Aragó corridor

Same length and width, east of the axis
0.317 km²

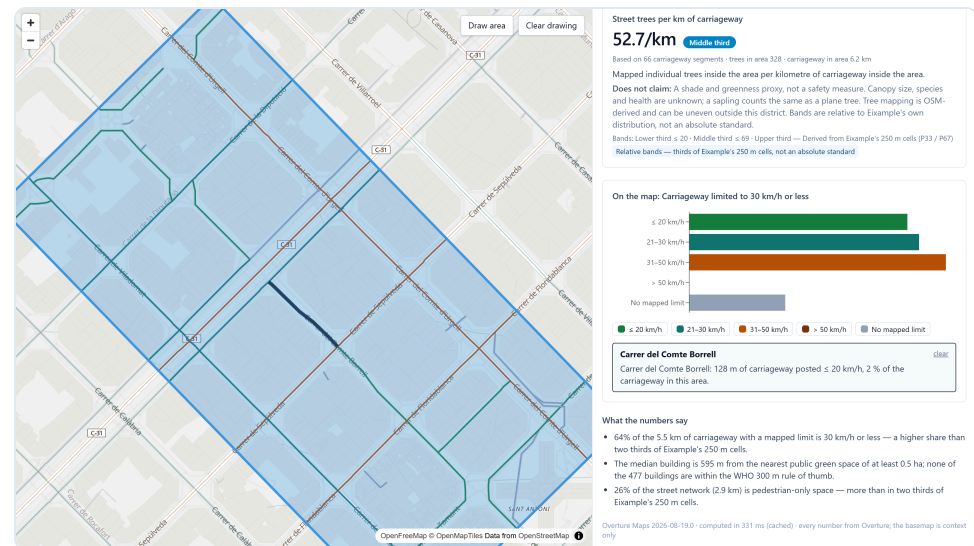
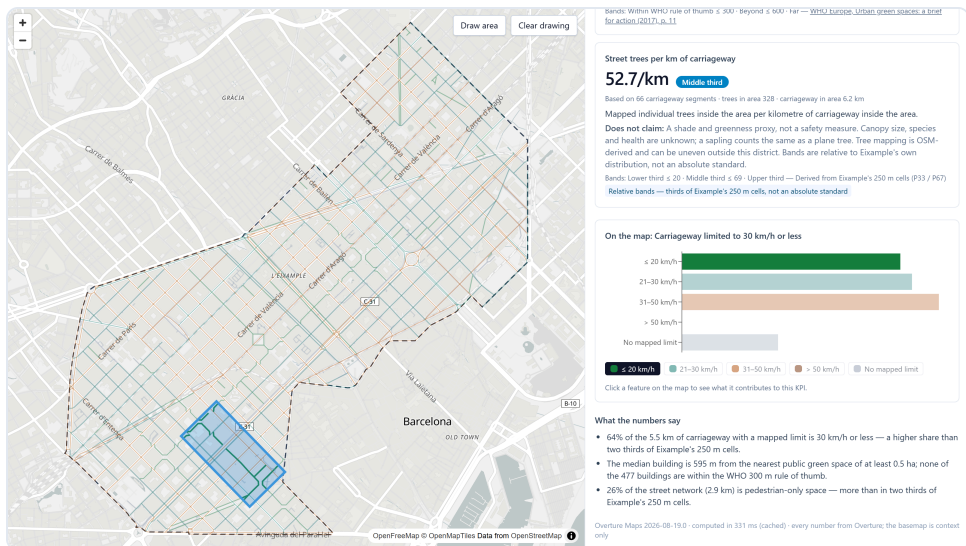
	A	B
≤ 30 km/h	63.6 %	47.7 %
Crossings / km	20.9	31.2
Pedestrian-only	26.0 %	16.7 %
Green p50	595 m	151 m
Trees / km	52.7	50.3



The intervention is in the data: Comte Borrell and Consell de Cent are **living_street** at 10 km/h.

One state, read both ways

A dashboard category highlights the map; a map feature explains its contribution



Dashboard → map. Clicking ≤ 20 km/h lights those streets and dims the rest — in area A, the living-street axes Comte Borrell and Consell de Cent. No refetch.

Map → dashboard. Comte Borrell: 128 m posted ≤ 20 km/h, 2 % of the carriageway in this area — computed from the loaded layer.

`selectFeature` → `describeContribution` — one pure rule per KPI

`toggleCategory` → `emphasisExpression` → `setPaintProperty`

Map and dashboard never import each other: they meet in one store that holds selection only.

What the numbers do not claim

The same comparison, read honestly

A "loses" two of five

Crossings — the meaning inverts. Shared space at 10 km/h is crossable along its whole length, so marked crossings are removed. This KPI **falls** where a planner succeeded.

Green — structurally blind. The Consell de Cent axis is *linear* greening; a "≥ 0.5 ha polygon within 300 m" rule cannot see it. B's 151 m is Sagrada Família's gardens.

So there is no composite score

Every card carries its own **does not claim** line, next to the number. Adding the five into one index would destroy exactly the caveats that make them usable.

Pedestrian crossings per km of carriageway

20.9/km Middle third

Based on 66 carriageway segments · crossings in area 130 · carriageway in area 6.2 km

Mapped pedestrian crossing points inside the area per kilometre of carriageway inside the area.

Does not claim: A crossing point carries no quality: signalised, raised, marked or lit are not known. High density does not mean safe crossing, and a pedestrianised area with no carriageway has nothing to cross and reads as no data, not as zero. In shared-space or living streets, fewer marked crossings can mean the whole street has become crossable; a low reading there is not a deficit. Bands are relative to Eixample's own distribution, not an absolute standard.

Bands: Lower third ≤ 19 · Middle third ≤ 26 · Upper third — Derived from Eixample's 250 m cells (P33 / P67)

Relative bands — thirds of Eixample's 250 m cells, not an absolute standard

Where every threshold comes from

	Threshold	Source
Cited	30 km/h urban limit	RD 970/2020, art. 50.1 RGC: "30 km/h en vías de un único carril por sentido"
Cited	300 m, 0.5 ha	WHO Europe 2017, p. 11: "public green spaces of at least 0.5–1 hectare within 300 metres"
Derived	42/60, 19/26, 14/24, 20/69	P33 / P67 across Eixample's 250 m cells
Chosen	600 m green band	Double the WHO distance
Chosen	0.5 km cell minimum	One segment would decide a cell's value
Chosen	75 % coverage warning	"A clear majority" — changes no value, only a warning

Derived bands are labelled **Lower / Middle / Upper third** and marked *relative to Eixample* on the card. Any evaluative word on a percentile cut would overclaim.

Both citations were opened and quoted from the source text, not from memory. `make derive-bands` regenerates the tertiles.

Three times, the data measured the mappers

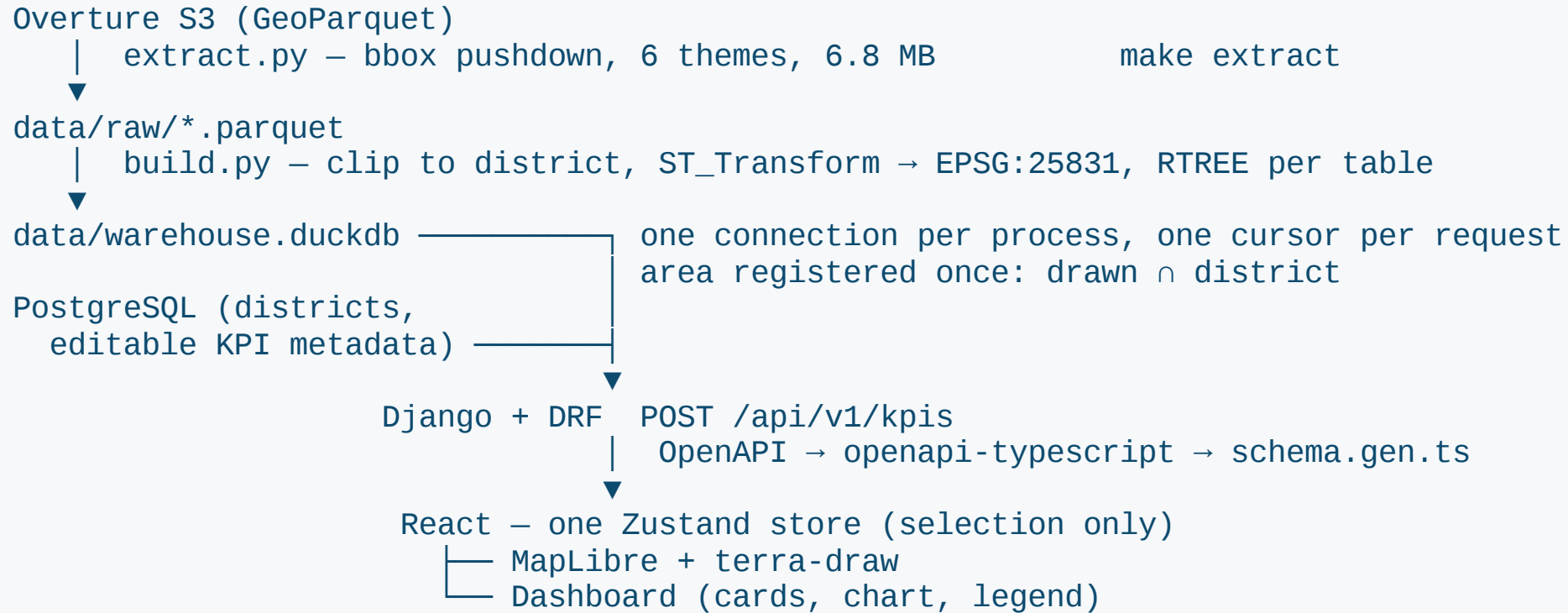
Street lamps — rejected. 236 in 7.5 km², 65 % of them in two 1-km cells; one full cell with 58.9 km of road has none. The KPI would have scored survey effort and called it "dark streets".

Land-use mix — rejected. Land use covers 29.7 % of the district, 61 % of that one class. An entropy over it measures coverage.

Pedestrian share — caught in build. v1 read 52 % district-wide and 49.9–55.2 % across cells: suspiciously flat. Sidewalk km per carriageway km ranged 0.68–1.54 (median 1.09) where a fully mapped grid gives ~2. **The variance was OSM sidewalk coverage.** Excluding sidewalks: **20.6 %**, spanning 0–71 % across cells, and it now measures something a planner can change.

Transit was rejected for the opposite reason: clean data, no variance — 100 % of buildings within 300 m of a stop.

Architecture



Numbers live **once**, in TanStack Query keyed by the area. The store holds selection.

Decisions, and what they beat

Chosen	Rejected	Why
DuckDB + GeoParquet	PostGIS	The extraction path <i>is</i> GeoParquet: one engine end to end, no load step
REST + OpenAPI	GraphQL	Generated client types for a fraction of the setup
Postgres for metadata only	All of it, or none	Bands and citations become rows a planner can audit
Prepared extract, public data repo	Commit it, or hit S3	Script committed, data not; running in 15 minutes
OpenFreeMap basemap	No basemap	Cartographic context only; no KPI reads it

```
WITH clipped AS (
    SELECT max_speed_kmh,
           CASE WHEN ST_CoveredBy(geom_m, area_m())
                THEN length_m           -- wholly inside
                ELSE ST_Length(
                    ST_Intersection(geom_m, area_m()))
           END AS len_m
    FROM segments
    WHERE is_carriageway
          AND ST_Intersects(geom_m, area_m())
)
SELECT 100.0 * sum(len_m) FILTER (WHERE max_speed_kmh <= 30)
       / nullif(sum(len_m) FILTER (WHERE max_speed_kmh IS NOT NULL), 0)
FROM clipped;           -- ...breakdown and context omitted
```

Area binding	Plan	District
Temp-table join	SPATIAL_JOIN	13.5 ms
Inlined WKT	RTREE	4.6 ms
Prepared ?	RTREE	5.6 ms
Macro over SET VARIABLE	RTREE	3.6 ms

Measured with `EXPLAIN`: DuckDB uses the R-tree only when the predicate argument is **constant at plan time** — hence a macro over a session variable, registered **once per request**.

No Python loop over features — and the denominator is visible in the query.

Four bugs that taught something

Axis order. `ST_Transform` to EPSG:25831 put Barcelona 4 700 km away — EPSG:4326 is officially (lat, lon) and DuckDB honours it. → `always_xy := true` on every transform, and a repo rule so it cannot come back.

Drawn $\cap$ district. A box drawn *around* the district read 51.3 %, the district 50.7 %. Segments touching the district are stored whole, so clipping to the raw drawing counted their outside halves. → clip once in the resolver; pinned by a test that draws a containing box.

`list()` ignores a CTE's `ORDER BY`. Chart bars reordered between identical requests when two classes tied. → sort inside the aggregate.

DuckDB caches instances per path. After `make load-data`, reopening returned the *old* file silently. → drain cursors, close, then reopen.

What breaks at 10×

Payload. The district ships 5.1 MB of GeoJSON (0.98 MB gzipped). At 10× districts or 10× zoom, that becomes vector tiles: static geometry served once, per-request ids and values joined with `feature-state`.

Computation. Clipping per request is right at 7.5 km². At city scale, precompute each KPI on an H3 grid and aggregate cells for large polygons, clipping only the boundary ring.

Storage. One DuckDB file → Parquet partitioned by district and theme, with the file the query engine reads chosen by the area.

Cache. Honest limit: drawn polygons essentially never repeat, so the cache serves the district and repeats. Redis moves it off-process; the real win is the H3 precompute above.

Concurrency. `runserver` → gunicorn workers, each with its own read-only connection and its own cursor per request. The connection layer is already shaped for that.

Latency. Past ~2 s a synchronous request is wrong: job queue, return KPI values first, stream layers after. Today the district is 2.4 s cold and 0.4 s cached; area A is 0.6 s and 0.04 s.

Another week

I would add

Compare two areas side by side — the A/B table is the most useful thing in this deck and the product cannot do it yet.

A second district, to find which of these caveats are Eixample's and which are Overture's.

Per-feature confidence from Overture's source metadata, so "unmapped" and "absent" can be told apart on the map rather than only in a caveat.

I would throw away

The five-KPI grid as a layout. One KPI at a time with its caveat and its map reads better than five cards competing.

street_tree_density — it is the weakest of the five for a safety reader and exists mostly to prove a second density.

The context strings in SQL. They belong in the registry with the labels; putting them in the query file made three files change when one label did.